

Final Project: The "Full-Stack Concepts" Web Application MVP

Activity Overview: Working in groups of four (4), you will research, design, document, and develop a functional "Minimum Viable Product" (MVP) of a web application. Your project must serve a specific purpose (e.g., a Library Management System, a Personal Finance Tracker, a Media Player, or a Small Business Inventory System).

The core requirement is that every technical decision—from your layout to your data handling—must be justified using the concepts from our 8 lecture modules.

1. General Instructions

- Group Size: 4 members.
- Deliverables:
 1. Project Files: A folder containing all files such as index.html, styles.css, script.js, and an assets/ folder.
 2. Documentation: A comprehensive PDF report (11pt Arial, 1.5 spacing).
 3. Final Presentation: A 10-minute presentation with a live demo.
- Naming Convention: Group#_ProjectTitle_Finals.zip
- Firm Rule: Your CSS must be in an external stylesheet, and your JavaScript must be in an external file to ensure maintainability.

2. Activity Phases

Phase 1: Research & Requirements (Documentation)

Before writing a single line of code, your group must:

- Background Research: Identify your website type. Find two existing professional examples. Document their functions and how your version will differ or improve upon them.
- Functional Requirements: What must the site do? (e.g., "The system shall calculate overtime pay using JavaScript logic.")
- Non-Functional Requirements: Performance, Responsiveness, and Visual Hierarchy, etc.

Phase 2: Design & The Cascade (CSS)

- Layout Strategy: You must implement a Grid for the main page structure and Flexbox for components like navigation bars or cards (refer to CSS Layout slides).
- The Box Model: Document how you handled overflow and used padding/margin to create white space (Box Model slides).
- Specificity & Inheritance: In your documentation, provide one example from your code where you had to use a Contextual Selector or ID Selector to overcome CSS Specificity (The

Cascade slides).

Phase 3: Media Optimization (Web Media)

- All images must be optimized. You must include at least one SVG for a logo and use WebP/JPEG for photos.
- If you include audio/video, you must provide multiple sources for cross-browser compatibility.
- Use HSL for your color scheme to ensure easy manipulation of "Hover" states.

Phase 4: System Logic (JavaScript)

- Data Handling: Use at least one Reference Type (Array or Object) to store data (JS Fundamentals, Slide 26).
- Logic: Implement a feature that uses Conditional Logic (if/else or switch) and a Loop (for/while) to process and display data dynamically.
- DOM Output: Use console.log() for debugging and document.write() or element manipulation to show results to the user.

3. Tools and Software

- VS Code: Your primary Integrated Development Environment (IDE).
- Chrome/Edge DevTools: Use the Lighthouse tool to check your performance and the Mobile Toggle to test your Media Queries (Responsive Design slides).
- Squoosh.app: For optimizing your raster assets.
- Canva/Figma: For initial UI prototyping and designing your presentation slides.

4. Notes

- On Critical Thinking: Do not just use a <div> for everything. Ask why a Grid is better for your gallery than a Float.
- On Logic: If your website is a Payroll System, use a switch statement for different employee levels. This shows you understand Conditional Logic.
- On Responsiveness: Your site must be usable on a 320px width phone. Ensure your Viewport Meta Tag is correctly set.

5. Reflection (To be included at the end of the report)

- As a group, reflect on the following: "How did understanding the Client-Server model change your perspective on what happens when a user clicks a button on your site, and what was the most challenging 'Cascade' conflict you solved during development?"

6. Criteria for Grading

Criterion	Excellent (20-18)	Satisfactory (17-14)	Needs Improvement (13- 0)	Points
Research & Documentation	Professional research with clear functional/non functional requirements and lecture integration.	Complete documentation but lacks deep analysis of existing competitors.	Incomplete documentation or fails to link lecture concepts.	/20
CSS Layout & Styling	Advanced use of Grid/Flexbox. Correct use of rem/em units and HSL colors.	Standard layout. Some misuse of units or absolute positioning where relative was needed.	Broken layout. No responsive design elements (Media Queries).	/20
JS Logic & Implementation	Flawless use of loops, arrays, and conditionals to solve a real-world problem.	Functional JS logic but contains minor errors or inefficient code (e.g., redundant if statements).	JS is non functional or extremely basic (e.g., only an alert).	/20
Media & Asset Management	All assets are optimized (WebP/SVG) and correctly implemented for cross-browser use.	Most assets optimized; some high-resolution files slowing the page.	No optimization; incorrect file formats used for the purpose.	/20
Collaboration & Presentation	Clear group synergy. Live demo works perfectly. Documentation is polished.	Good presentation but minor bugs during the live demo.	Poorly prepared presentation or lack of participation from all members.	/20
Total				/100